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RESEARCH

A novel fuzzy system design for data-adaptive secret key generation and robust audio encryption

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Abstract Fuzzy logic is a computing paradigm that allows systems to handle imprecise or uncertain data, making them more adaptable and robust. The paper proposes the novel artificial intelligence fuzzy encryption scheme for multimedia data such as audio. With the onset of 5G and 6G development, there is a broader growth of mobile users and bandwidth-hungry wireless applications. This, in turn, mandates ultra-low latency data delivery and utmost end-to-end security as the primary concerns for wireless network designers. The constant evolution of AI and its application in cyber security is a double-edged sword, as both attackers and defenders leverage their power to outmanoeuvre each other. As computational power increases, cryptographic algorithms may need unpredictable keys to maintain security against potential attacks. So, the work proposes that the key is generated by the fuzzy transformation process, making it crucial for the decryption process. In addition, the fuzzy composition can be applied in privacy-preserving computations and secure computation over encrypted data with some degree of fuzziness, protecting individual data while still allowing meaningful results. The proposed scheme is validated against security metrics and NIST tests. It exhibits enhancements such as entropy around 14, scrambling degree equal to unity, adjacent correlation coefficient values closer to 0 at the range of 10^{-5}, and a large key space of 2^{997}.

Keywords Fuzzy logic · Fuzzy inference system (FIS) · Artificial intelligence fuzzy encryption scheme (AIFES) · Key generation · Audio encryption

1 Introduction

The Fuzzy system is currently invading several application areas due to its ability to cater to uncertain decision outcomes. This is required in a real-time environment with the unsuitability of static mathematical decision-making. Fuzzy inferences are observed to be more adaptive for uncertain problems. Fuzzy rules and inference systems are widely adopted, especially for detecting malicious network activities and for accurate and lightweight decision-making [1]. Wireless communication systems originated several years ago and have strode from first-generation cellular telecommunication (1G) to the successor

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6G, under development with the associated technical advancements. The fifth-generation mobile communication systems primarily focused on improving data rate, spectral efficiency, reliability, and communication security. This paved the trail to applicability in recent critical applications such as the Internet of Things, health care, device-to-device communication, Machine-to-machine communication, and vehicular communication. The improvement in data rate and spectral efficiency has paved the way for prodigious growth in the number of users with seamless connectivity. Apart from the users, the information composition has evolved from voice to varied multimedia for various applications [2, 3]. The successor 6G primarily targets extending the bandwidth to other spectra, such as Terahertz/mm-Wave, utilising the essence of Artificial intelligence techniques in communication systems and robust security at the different protocol stack layers [4].

One of the key technologies of 5G, enhanced mobile broadband, is quite bandwidth-hungry and aims to provide the best user experience in terms of HD multimedia streaming and augmented/virtual reality. These applications would require the security measures associated with the respective applications. Another key technology, massive machine-type communications, demands lightweight encryption techniques. Ultra-reliable low latency communications expects low latency security techniques for real-time applications; with the onset of new applications due to the attractive features of 5G, the sensitive data associated with the application must be protected through multi-level authentication and privacy preservation [5]. Openness in the 5G network significantly contributes to the intruder eavesdropping the message between intended parties. Cryptography is best suited for encrypting the message in transit to protect the data from eavesdropping [6]. The security perspective is critical for 5G systems in IoT due to the mobile, distributed nature and openness to the internet [7]. The 5G and upcoming generation of communication systems would be demanding the complexity of the human nervous system in terms of secured infrastructure and adaptive behaviour. This has led to the application of artificial intelligence techniques to secure communication.

In the literature, chaotic artificial neural networks and fuzzy encryption schemes are experimented with to build a cryptographic model against various security attacks [8]. Fuzzy systems are preferred as a security tool among the AI techniques available for resource-deprived wireless networks [9]. This is mainly attributed to its lesser computational complexity and capability to solve problems involving uncertainties.

2 Related works

In the case of IoT involving wireless sensor networks that incorporate 5G and upcoming 6G, the interaction between the nodes, user devices, and servers is prone to tampering and eavesdropping, considering the recent advancements and mobility in the micro-scale interconnected world. In multimedia communication, audio is frequently transacted and bulky with highly harmonious consecutive values. Thus, the securing technique must eliminate the bulky nature of the method and successively encrypt the audio amplitude values statistically uncorrelated and undecipherable. While achieving this, the algorithm's energy and time consumption must be considered for real-time applicability in energy-constrained nodes of wireless networks.

The fuzzy rule-based inference system was used as an efficient tool to select the intermittent nodes in a distributed network based on chosen criteria. This enables the effective elimination of the suspicious nodes [10]. However, the number of attack attempts alone is considered a securing criterion for decision. This imposes the overhead of maintaining attack monitoring system results. This is extended to fuzzy trust value computation for peer-to-peer mobile user communication. This shows that a fuzzy inference system is suitable for securing communication techniques [11]. Similar work has been done for cloud service providers' trust evaluation [12]. Though the trust value could be evaluated and the peer is chosen as the next step to protect the data in transit, the fuzzy systems could be further explored to encrypt and authenticate users. On further survey of encryption and key generation techniques specifically for audio data chosen, the Linear Predictive Coding technique is adopted for voice encryption. This succeeds in achieving the desired rate but suffers from the distorted encryption and the overhead of codebook design [13].

Bio-inspired DNA encoding has been experimented with for audio data blocks with distinct keys for each

block [14]. DNA encoding maps every two bits of the audio sample to the A/T/G/C components, and each component is stored as 8-bit individual values, which is inefficient data. This results in overhead in the case of conventional computers. Though the distinct keys are utilised for each block, the initial key is the only one that contributes to the key sensitivity/keyspace, and the rest are derived from the previous key. Hence, it's prone to differential chosen-plaintext attacks.

Further, since the shuffling is accomplished within the tiny block through a swapping operation, the permutation would be fragile, especially considering the distribution of voice data [15]. Furthermore, for diffusion, the simple XOR-based operation is used in most existing works, which is prone to chosen plain text attack [16–19]. Also, in the case of the key generation for hashing-based authentication, the simple XOR-based operation is used in the existing work [20]. Due to the smaller public key size, the elliptic curve cryptography (ECC) is adopted for audio encryption [21]. Notably, the implementation complications in ECC are significant due to the standard curves involved. Deep neural networks such as autoencoder encode the image pixel values with uniform distribution constraints [22]. However, using a deep neural network to encrypt multimedia data adds an overhead of transmitting the weight matrices, bias vectors, and other network parameters to the receiver side. These are required to recover the original data. However, the neural network-based random sequence generation is adopted for the confusion stage of encryption [23].

Another prominent yet computationally efficient bio-inspired artificial intelligence technique, Fuzzy logic, is proposed in the literature to encrypt medical images for secure transmission over the channel [24]. Also, fuzzy logic effectively achieves the region of interest (ROI) extraction to perform watermarking in medical images [25]. The malicious, uncertain behaviour of the insiders could be well identified through fuzzy logic [26]. The security robustness is tightened by efficient data randomisation to break the correlation between the adjacent values. Chaotic cryptography has demonstrated appreciable performance for this purpose in the literature for the audio signal [27–29]. A logistic map is observed to outperform other preliminary chaotic maps in terms of cryptographic properties and lightweight in terms of complexity [30–35]. Considering the bigger size of the audio file, the number of operations performed should be less to achieve a good scrambling performance.

DWT-based encryption is attempted on audio signals [36]. The encryption and embedding combination will be complicated, but the key sensitivity won't be good. The three-level audio securities through the confusion in the frequency transform domain, namely DCT, IWT, and FFT, for equivalent diffusion in the spatial domain, followed by diffusion and confusion in the spatial domain using DWT and logistic map, respectively [37]. This work could not be preferred due to the imaginary terms and absence of energy compaction in FFT. In addition, the keys used to encrypt the plaintext are presumed to be shared on separate secure channels in advance, which isn't feasible in the present-day scenario. Combining DNA with chaotic maps for audio encryption would require a secure key exchange mechanism [38].

Further, using the preloaded keys longer isn't robust enough. Thus, the key will be generated dynamically based on the user input data [39]. This work at the concrete utilises complex SHA256, which causes considerable time consumption with the increase in the number of keys. Further, the randomness of keys generated depends on the uniqueness of user question/answer pairs, which is uncertain. Also, the key exchange in an open channel is prone to eavesdropping. A genetic algorithm-based secured key generation employs a fitness function to strengthen the keys [40]. However, the randomness of the key is enhanced by combining the DES and genetic algorithm. It should be noted that these two algorithms are computationally expensive individually. Further, the randomness is attributed to the random numbers generated as the initial population. The keys can be dynamically generated from the location attributes in the case of mobile devices in the network [41]. However, the seed source for key generation can be generalised for static and mobile devices. The deliberate chip error introduction allows for hardware authentication in IoT devices [42]. In the mobile cloud computing (MCC) context, garbled circuits and authenticated cryptographic computations can secure key management and enhance privacy [43]. Thus, the lightweight key generation approach to preserve privacy in edge environments can be explored.

In the proposed work, fuzzy composition-based encryption employs the pre-agreed compositions

chosen by the peers that authenticate the robustness against statistical attacks. The primary motivation behind adopting fuzzy logic to achieve ciphering is to analyse the data transformation through fuzzy compositions to perform adaptive image diffusion. Further, it doesn't involve multiple time-consuming iterations, making it best suited for efficiently transmitting speech signals over wireless networks. Unlike the image, the audio signal is large in volume and characterised by redundancy in amplitude values. Thus, in the proposed work, fuzzy-based encryption is manoeuvred for audio signals, along with enhancements in the fuzzy compositions chosen for tailoring the algorithm to fit the best performance of audio encryption. Multiple chaotic maps were considered for attaining the speech encryption multi-level and via multiple scrolls, which are many resources devouring [44, 45].

The logistic map with fewer operations exhibits better spatial correlation performance than the multi-map hybrid chaotic system. Thus, the proposed work employs the logistic map to confuse the fuzzy-diffused amplitude values. The DWT, followed by DCT, is used to achieve compression via scanning before encrypting the speech signal, where lossless compression could be achieved [46]. However, it is not realisable for resource-constrained 5G/6G wireless networks in current applications, and we couldn't expect a real-time response due to the communication complexity of DWT compared with DCT. Considering the resource constraints, the partial encryption of audio data is experimented with in literature to reduce the energy consumption of wireless nodes [44]. However, this work limits the size of data in each block for encryption and prioritisation based on an index that is not optimal in the case of an audio signal. An asymmetric Numeral System is used to attain encryption [46].

However, the results demonstrate the trade-off between the efficiency and the robustness of encryption. The randomness, however, depends on the binary pseudorandom generator and XOR operations. Thus, the keyspace and sensitivity would be very low. Compcrypt with encoding function evolution seems to outperform the techniques proposed in this work. But, it involves computationally inefficient re-computation of the whole encoding table for each symbol. Multi-dimensional chaotic maps and neural networks were employed to achieve key generation/encryption and resist brute-force attacks [48]. However, it does not address the lightweight requirement [49–52]. The fuzzy inference system (FIS) was experimented with to enhance the security parameters for IoT cloud data storage [51].

Thus, we must rely on complex multistage key generation techniques to generate a secure key from existing literature. These techniques are computationally intensive, and in most works, multiple methods are combined to attain the intended robustness of the key. This motivated us to experiment with a Fuzzy Inference System (FIS) to generate the dynamic key from the peer's input. To our knowledge, no literature uses FIS to generate audio statistical features. Thus, in the proposed work, the dynamic key is generated from the inherent features of the audio stream through FIS design. The keys are concealed by transacting only the significant audio features to the counterpart to generate keys symmetrically. Considering the energy constraint and the robustness required, the less computationally complex fuzzy inference system could generate the needed keys. This is primarily known for dynamic modelling with the statistical features of the audio properties as inputs.

Further, we extended the Fuzzy system to encrypt the audio data. Though Fuzzy Logic is employed for encrypting an image in literature, unlike the image, the audio signal is large in volume and characterised by redundancy in amplitude values. In the literature, fuzzy systems have been successfully experimented with image data, demonstrating strong performance in terms of security and efficiency. This cross-domain effectiveness indicates that the proposed method is adaptable and robust. Provided its effectiveness in audio and image encryption, it holds great potential for application to video data, which combines both modalities. Thus, in the proposed work, fuzzy-based encryption is manoeuvred for audio signals, along with enhancements in the fuzzy compositions chosen for tailoring the algorithm to fit the best performance of audio encryption. In the proposed work, fuzzy composition-based encryption employs the pre-agreed compositions selected by the peers that authenticate the robustness against statistical attacks.

The primary motivation behind adopting fuzzy logic to achieve ciphering is to analyse the data transformation through fuzzy compositions to perform adaptive image diffusion. Further, it doesn't involve multiple time-consuming iterations, making it best

suited for efficiently transmitting speech signals over wireless networks.

1. Enhanced AIFES-based diffusion for ciphering audio
2. Non-iterative Fuzzy composition transformation is simple yet efficient
3. AIFES prevents the plain text attack due to the varied fuzzy compositions employed
4. A fuzzy inference-based key generator enables dynamic and adaptive audio attributes-based key generation
5. FIS-based key generation removes secure key exchange overhead as only audio attributes were required at peers to derive secret keys
6. The proven data/time saving and robustness validation of the proposed audio cryptosystem for wireless networks
7. The experiment demonstrates promising data/computational efficiency results and robust security against eavesdroppers.

3 Proposed methodology

Figure 1 represents the overall proposed work block diagram. The audio data are huge in size with closely correlated adjacent values. This demands artificial intelligence indulgence to transform the amplitude levels of audio to undecipherable noisy audio. The proposed approach generates the dynamic keys from the audio signal through the Fuzzy Inference System. The audio signal read is passed through enhanced fuzzy composition-based diffusion. The diffused amplitude values are weighted based on two additional categorical keys followed by fuzzy composition. The weighted values are normalised for uniform distribution by deducting average values of respective composition outcomes. Further, the spatial association between the adjacent values in diffused data is brought down through scrambling using the logistic map as in Fig. 1.

The proposed method consists of three phases.

- FIS aided adaptive key generation
- Key generation randomness test
- AIFES-governed audio encryption

3.1 FIS aided adaptive key generation

For the key generation purpose, the audio attributes set 'O' which comprises harmonic ratio, pitch, spectral entropy, spectral flatness, spectral flux, spectral roll-off point, spectral skewness, spectral slope, spectral spread, bark spectrum, spectral kurtosis, and amplitude values are considered [49]. Six inputs were expected to be fed to the key generator for a key generation.

The two audio attributes would be considered simultaneously for four split-up audio blocks. This would be repeated to compute the statistical metrics for all the audio attributes. This sequence generated would be fed as input to the key generator as in Fig. 2a. The predefined fuzzy rule set and activation functions shall be used for key generation. These rule sets and activation functions are already agreed upon between the counterparts via secure channels, as in Fig. 2b. The rule sets and activation functions will be loaded onto the peer devices during the initial registration. The first round will use the first combination of rule set and activation functions. For further rounds, the activation functions and rule sets will be chosen randomly by the peers based on the last round key values α and β. The respective regulations set and activation function combination chosen will be $\{(\alpha^*\beta) *10^4\% \upsilon\}$. The υ indicates the maximum number of rules set and activation function combinations available with the peers. Note that these keys are generated as dynamic keys during the encryption process. Thus, the chances of key retrieval and the intruder's rule sets/activation function identification will be infinitesimally small.

The following steps emphasise the impossibility of an intruder to decipher the secret key used,

1. High entropy seed selection:
   The unique seed values chosen for each session make the replay or prediction attacks ineffective.
2. Robust key generation:
   Without the knowledge of activation functions, fuzzy rule sets and seed value, the key generation is computationally infeasible. The initial rule set and activation functions loaded during registration are used for the first session. For further rounds, the rule set and activation functions are randomly chosen from a myriad of options available based on the last session dynamic keys. This adds an

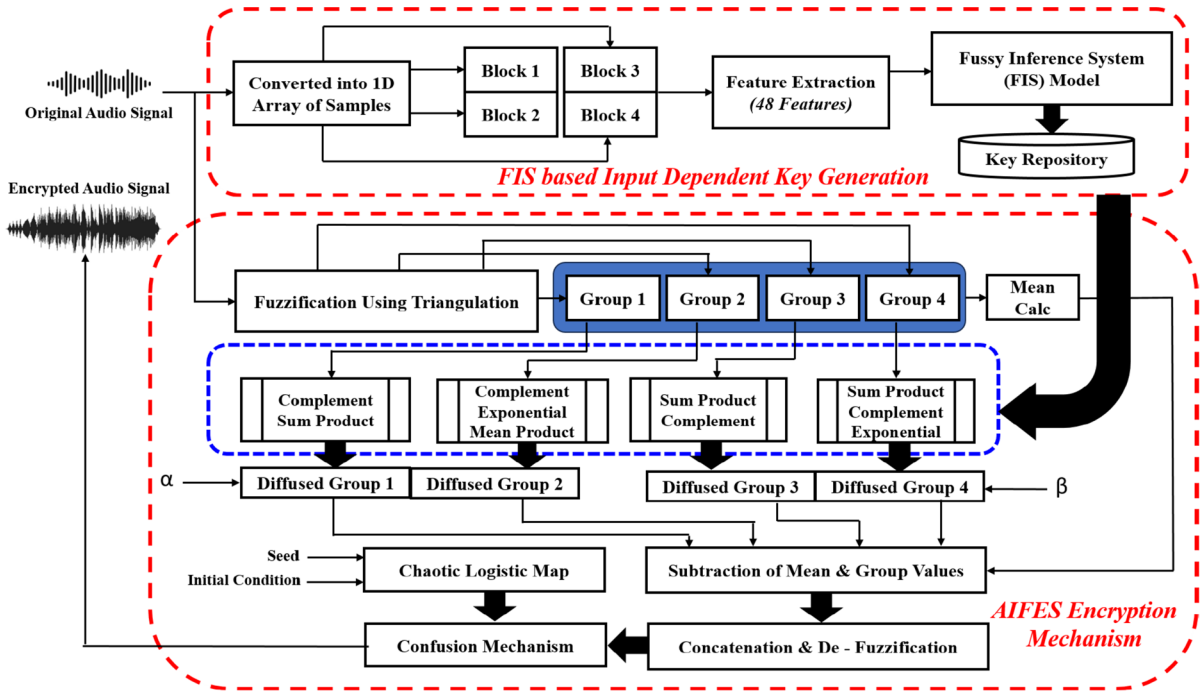
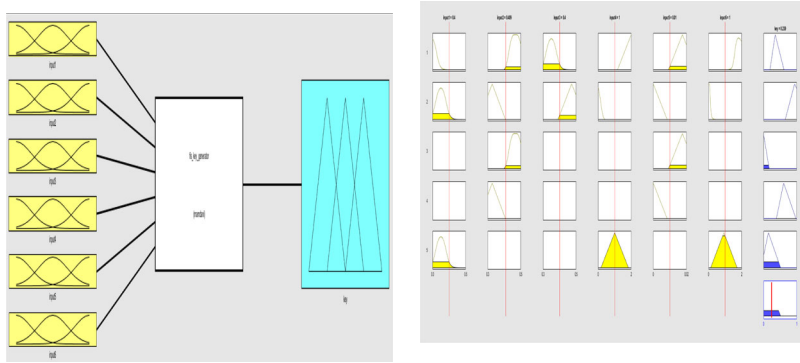


Fig. 1 Proposed multimedia cryptosystem

Fig. 2 **a** FIS model for secret key generation, **b** graphical representation of the fuzzy rule set considered



(a) (b)

extra layer of protection and prevents the intruder from deciphering the secure keys.

3. Strong encryption process:
   The fuzzy composition-based encryption scheme is performed as an additional layer of protection. In addition to six keys from the key generation block, two dynamic keys generated during the encryption process will be required to decipher the encrypted message.

Further, the proposed work allows independence in choosing the activation functions, the number of inputs, and the rule set used. This would result in diversified keys for the same audio file. Thus, the intruder in transit with the partial audio properties could not decipher the key without knowing the rule set and activation functions adopted in the proposed FIS model. The proposed key generation technique is depicted in algorithm 1. Where $A_v(.)$, $M_d(.)$, and $M_o(.)$ represent the mean, median and mode of the vectors. The audio inputs with low variance can potentially

limit the robustness of the encryption process. Thus, we have incorporated the statistical parameters corresponding to the audio signal's time and frequency domain features. Combining time-domain and frequency-domain features to arrive at statistical parameters will amplify the skewness of the input to FIS [54]. The additional randomness and resilience to key reconstruction attacks are introduced through the nonlinear transformation by FIS.

3.2 Key generation randomness test

To assess the uncertainty of the keys generated, the NIST test, which comprises 15 major statistical test suites, was performed [53]. The p-values indicate the level of preciseness of the random key generated. The p-value would be compared with the reference threshold value of 0.01 to decide the status of the respective test. The significant random digits of the key generated are converted into an equivalent binary stream and considered for testing. The higher p-value indicates the higher randomness in the key stream.

The p-value corresponding to the binary rank test is quite high, indicating that the sub-sequences of the whole keystream generated are disjointed. Hence, this would result in the elevated randomness of the keystream. The DFT test p-value closer to 0.8 indicates the randomness of the peaks in the frequency domain of the sequence. This validates the absence of the repeating patterns in the keystream. There were no template pattern matches as per the higher values of the template matching test. The key generation is inferred to be complex with a very high linear complexity test p-value value. The average entropy test p-value is at a peak of 0.98, indicating the proposed scheme's randomness as in Table 1.

3.3 AIFES-governed audio encryption

Figure 1 presents the proposed AIFES audio encryption scheme. The audio signal is considered for Fuzzy diffusion. Initially, the data is divided into fixed-sized blocks. The size of the blocks is up to the user's choice since the ultimate number of computations remains the same. The blocks are clustered into four different groups. Each group is operated with the user's four different compositions. The user can choose the composition based on complement, exponential, sum, mean, product, and so on, which are reversible. Thus, the composition is chosen and pre-shared between the peers and the keys. The four compositions selected were as follows:

Table 1 NIST statistical test suite results for comparison with existing work

Type of test	P-value (proposed)	P-value [30]
Frequency test within a block	0.11185	0.035174
Longest run of ones in a block	0.70424	0.213309
Binary matrix rank test	0.69372	0.213309
Discrete Fourier transform (spectral) test	0.838905	0.350485
Non-overlapping template matching test	0.862828	0.534146
Overlapping template matching test	0.488416	0.350485
Linear complexity test	0.919689	0.350485
Serial test	0.593506	0.350485
	0.844766	0.534146
Approximate entropy test	0.989322	0.350485

Algorithm 1 Dynamic data-drive fuzzy Inference based secret key generation

Input : Audio Data Inp of size $M \times 2$
Output: Secret keys K_k, $\forall k \in [1, 6]$
$A[1 \times N] \leftarrow Reshape(Inp_{M\times 2})_{1\times(M\times 2)}$,
$B[4 \times (1 \times \frac{N}{4})] \leftarrow Split(A)$,
$S_t \leftarrow \{\}$
for $j \leftarrow 1$ *to* 12 **do**
  for $i \leftarrow 1$ *to* 4 **do**
    $[D1, D2] \leftarrow O(B[i])_{j,j+1}$
    $D1_stat \leftarrow A_v(A_v(D1), M_d(D1), M_o(D1))$
    $D2_stat \leftarrow A_v(A_v(D2), M_d(D2), M_o(D2))$
    $S_t \leftarrow \{S_t, D1\ stat, D2_stat\}$
  end for
  $j \leftarrow j + 2$
end for
for $k \leftarrow 1$ *to* 8 **do**
  $K_k \leftarrow FIS(S_t\{1 \times 6\})$
end for
return K_k, $\forall k \in [1, 6]$

$$FuzzyComp1 = C\{\begin{matrix} A & B \\ C & D \end{matrix} \times \begin{matrix} A & B \\ C & D \end{matrix}\} \quad (1)$$

$$FuzzyComp2 = C\{\frac{e^{\begin{matrix} A & B \\ C & D \end{matrix} \times \begin{matrix} A & B \\ C & D \end{matrix}}}{2}\} \quad (2)$$

$$FuzzyComp3 = \{\begin{matrix} C(A) & C(B) \\ C(C) & C(D) \end{matrix} \times \begin{matrix} C(A) & C(B) \\ C(C) & C(D) \end{matrix}\} \quad (3)$$

$$FuzzyComp4 = \{\begin{matrix} f(A) & f(B) \\ f(C) & f(D) \end{matrix} \times \begin{matrix} f(A) & f(B) \\ f(C) & f(D) \end{matrix}\} \quad (4)$$

where f(.) is the C(exp(.)). Once the composition results are obtained, the wholesome result is divided into two groups. The two additional keys scale each group. The scaled groups are normalised by subtracting the respective mean values. These two groups are joined, and the final diffused result is formed. Thus, the fuzzy logic-based diffusion technique adopted is as in Algorithm 3

Algorithm 2 AIFES-based audio diffusion

Input : Compressed Audio samples A_{comp} of size $1 \times N_{comp}$
Output: Encrypted Audio of size $1 \times N_{comp}$
$A_F \leftarrow \frac{A_{comp} - min(A_{comp})}{max(A_{comp}) - min(A_{comp})}$
$A_{F\,limits} \leftarrow [max(A_{comp}), min(A_{comp})]$
$K_{F[2\times3]} \leftarrow K_k$
$C_F \leftarrow [(1), (2), (3), (4)]$
$A_F \leftarrow Split(A_F)$
$G_{F[4\times(2\times2blocks)]} \leftarrow Cluster(A_F)$
$G_C[G_F[i]] \leftarrow GroupAssignment$
for $i \leftarrow 1$ *to* $\frac{Ncomp}{4}$ **do**
  $\circ \leftarrow C[G_C[G_F[i]]]$
  $C_F[i]_{2\times2} \leftarrow G_F[i] \circ K_{F[1:2,1:2]}$
end for
$C_N \leftarrow Norm(C_F, [-1, 1])$
$G_{F1} \leftarrow G_F[:, 1 : \frac{Ncomp}{2}]$
$G_{F2} \leftarrow G_F[:, \frac{Ncomp}{2} + 1 : N_{comp}]$
$G_{F1} \leftarrow G_{F1} \times K_F[3, 1]$
$G_{F2} \leftarrow G_{F2} \times K_F[3.2]$
$G_{F1} \leftarrow G_{F1} - mean(G_{F1})$
$G_{F2} \leftarrow G_{F2} - mean(G_{F2})$
$C_{Fin} \leftarrow [G_{F1};\ G_{F2}]$
$A_{F\,limits} \leftarrow [A_{F\,limits};\ [mean(G_{F1}), mean(G_{F2})]]$

return $C_{F\,in}$

The audio amplitude values are transformed through the fuzzy compositions without XOR operations. The respective amplitude values are real numbers with the probability of deciphering close to infinity without the keys. Final ciphered data need to be randomised to eradicate the spatial correlation. Hence, the Logistic map-based chaotic map is employed. The initial seed value and the parameter μ are identified as keys for randomisation. The slight change in these values would result in a huge difference in the chaotic sequence generated.

$$[X_{n\,sorted},\ X_{n\,sort\,index}] = sort(X_n,\ ascend) \tag{5}$$

The chaotic sequence generated is sorted in increasing order, and the respective indices are used as index maps to confuse the diffused audio signal. The decryption necessitates the reversal of confusion utilising the Logistic map-based index generation. The reverse fuzzy decomposition is then applied to retrieve the audio signal.

To validate the proposed scheme, nine different audio files with unique spatial features and statistical distribution were considered as in Fig. 3. Various security analysis metrics were considered, such as correlation coefficient, entropy, scrambling degree, PSNR, and NSCR. Further, the mean square error and correlation coefficient were obtained to ensure the data recovery performance. The proposed work is compared with the pre-existing schemes.

3.4 Correlation analysis

The correlation coefficient value determines the closeness between the adjacent samples in the data. Generally, the audio data stream is characterised by closely correlated subsequent sample values. The ideal encryption is characterised by dispersing the values such that the correlation coefficient is closer to zero. This ensures that the adjacent samples aren't related and challenges the intruder from deriving the subsequent values from the previous value [29]. From

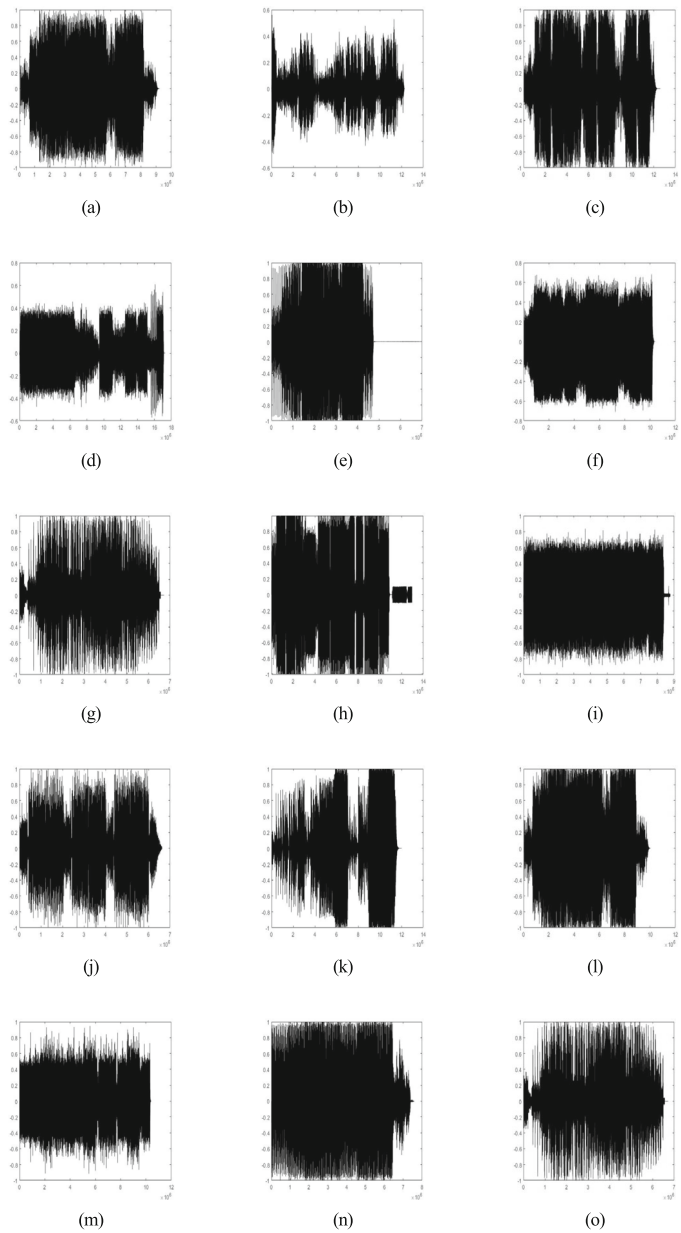


Fig. 3 Test audio input files of different spatial distributions considered for analysing the proposed methodology

Fig. 4a, audio files exhibit high correlation values between adjacent samples as the correlation coefficients are closer. The proposed method seems to have brought down the correlation values closer to zero; hence, the subsequent values won't let the intruders decipher the confidential data. The logistic map utilised for scrambling appears to confuse the audio samples effectively while combined with the fuzzy-based diffusion of audio samples.

3.5 Entropy analysis

Entropy provides the measure of randomness in values distributed throughout the audio stream. The higher the uniformity in the distribution of audio samples amplitude, the more the entropy measure of audio data [29]. The audio data acquired takes values from -1 to 1 with the 4-digit precision post the decimal points. Thus, the number of histogram bins ranges up to 20,001, which can attain the corresponding maximum

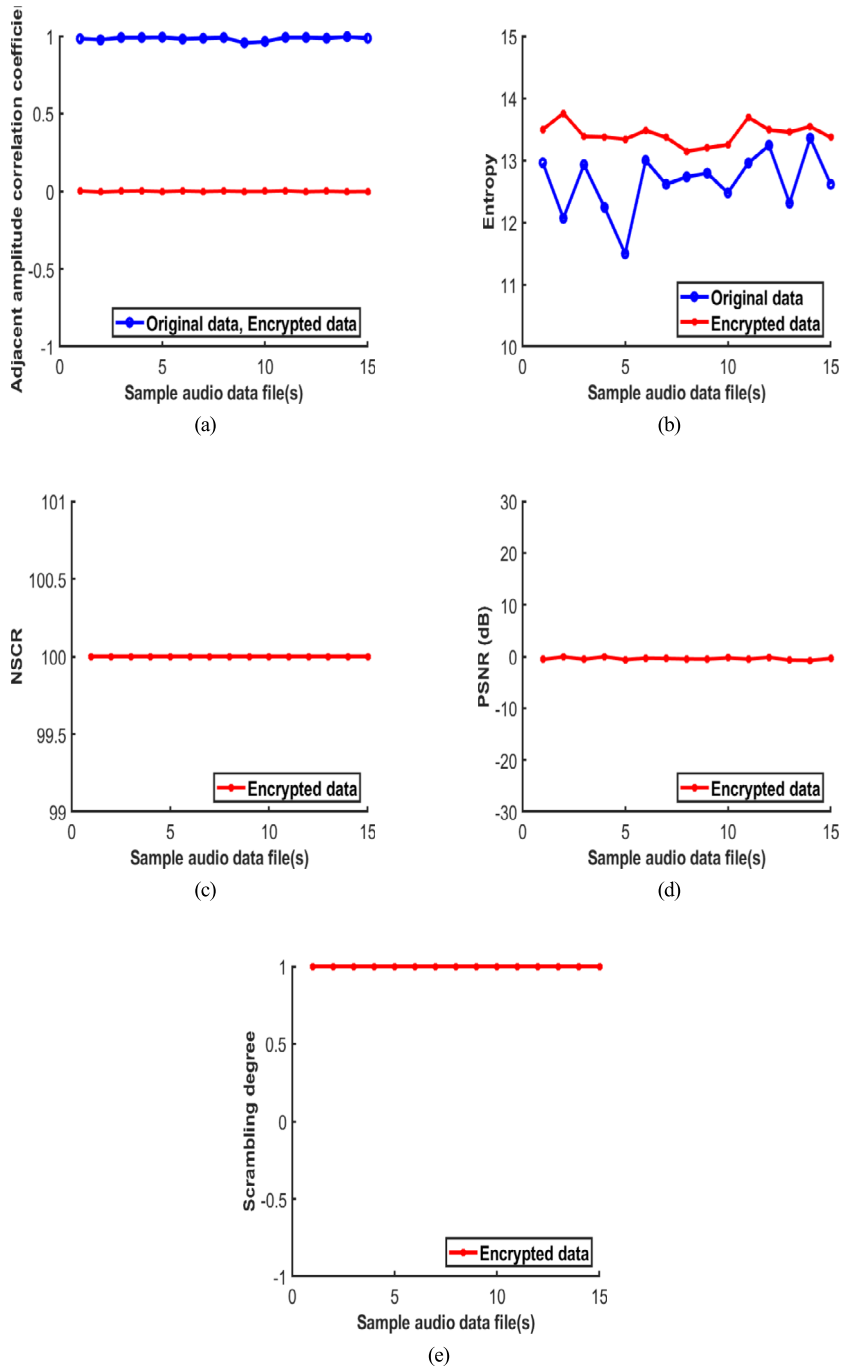


Fig. 4 **a** Comparison of adjacent amplitude correlation coefficients between origi- nal and encrypted audio. **b** Comparison of entropy in original and encrypted audio. **c** Number of Samples Changes Rate between original and encrypted audio. **d** Peak signal ratio of encrypted audio with respect to original audio. **e** Scrambling degree of proposed encryption technique ultimate key sensitivity which could change the cipher data obtained to be entirely different through the proposed work which secures the audio from brute force attack

entropy of between 14 and 15. Figure 4b shows that the entropy value has increased considerably post-encryption. This depicts that the Fuzzy diffusion, though non-iterative, increases the randomness of the audio encrypted. This is closer to the maximum value and assures protection from Brute force attacks.

3.6 Number of samples changes rate analysis

The number of samples changes rate indicates the percentage of total audio samples changed in encrypted audio concerning the original audio. The ideal encryption method should achieve a higher value of 100%. This indicates that all the values were changed during ciphering, which resists the man-in-the-middle and differential attacks [29]. NSCR values are observed to be 100% for all the test samples presented in Fig. 4c. This indicates that the fuzzy-based diffusion technique could change all the audio input samples into different ones in the proposed work.

3.7 Error metric analysis

Error metric PSNR indicates the degree of distortion in the ciphered audio concerning the audio data. The lower value of PSNR shows that the encrypted audio could be perceived as noise. The PSNR values are 0 dB for all the inputs presented, as in Fig. 4d.

3.8 Scrambling degree

The scrambling degree indicates the confusion in encrypted audio for the original audio [29]. The scrambling degree is obtained for all the audio samples presented in the proposed encryption scheme, as shown in Fig. 4e. The scrambling degree value of one indicates the complete scrambling of the encrypted audio against the original audio. This shows the exceptional confusion performance of the logistic map when integrated with the fuzzy diffusion.

3.9 Recovery performance analysis

Error metrics such as Mean Squared Error and respective peak signal to noise ratio between the original and recovered audio indicate the calibre of the proposed scheme regarding encryption credibility. The MSE value closer to zero and correspondingly high PSNR marks the ability of the proposed encryption scheme to preserve the quality of data transmitted [17]. The mean squared error value is much closer to zero for the recovered audio than the original audio. Correspondingly, the PSNR value is high, closer to 65 dB, and the correlation coefficient is almost one for all the audio data in Fig. 5. This depicts that the proposed work could recover the audio transmitted without much loss in transit.

3.10 Key sensitivity

Key sensitivity indicates the minute change in key value, which could completely change the final encrypted data. The highly sensitive key is generated adaptively from the audio data with the FIS assistance in the proposed work. For example, the key K_F [1, 1] is changed from 0.000001200000006 to 0.000001200000003 in the experiment to analyse the key sensitivity. This minor modification of 10^{-15} could achieve the Number of Samples Changes Rate of 99.9877%. This shows the ultimate key sensitivity, which could change the cipher data obtained to be entirely different through the proposed work, which secures the audio from brute force attack.

3.11 Keyspace

A total of six keys (K_F [2 × 3]) were used for fuzzy diffusion. The two keys were used for chaotic confusion (X_n, μ). These were generated using the proposed FIS key generation mechanism. Apart from these, ten fuzzification/normalisation ($A_{F\ limits}$[2 × 5]) limit values, the dynamic keys generated during the encryption scheme, are required on the receiver side to decipher the data. Each key is of 15-digit precision after the decimal point. Thus, the total actual keyspace goes to more than $(10^{15})^{20} = \cong 2^{997}$. Fuzzy composition-based encryption involves 6 different user key inputs and two adaptive keys derived from the intermediate data during the encryption process.

3.12 Computational time analysis

The primary essence of the proposed work is the real-time applicability in the current 5G integrated wireless energy-constrained networks. Thus, the computational time consumption of the proposed work is expected to be very low. Assessments are performed on a system with Intel(R) Core(TM) i3-4000M CPU @ 2.4 GHz, 4 GB RAM equipped with the MATLABR2019b environment. From Fig. 6, the lightweight diffusion and confusion consumed the meagre time of 1.749 ms and 0.594 ms on average for processing 32 KB of audio data. Thus, the total average time consumption is 2.343 ms for the proposed cryptosystem.

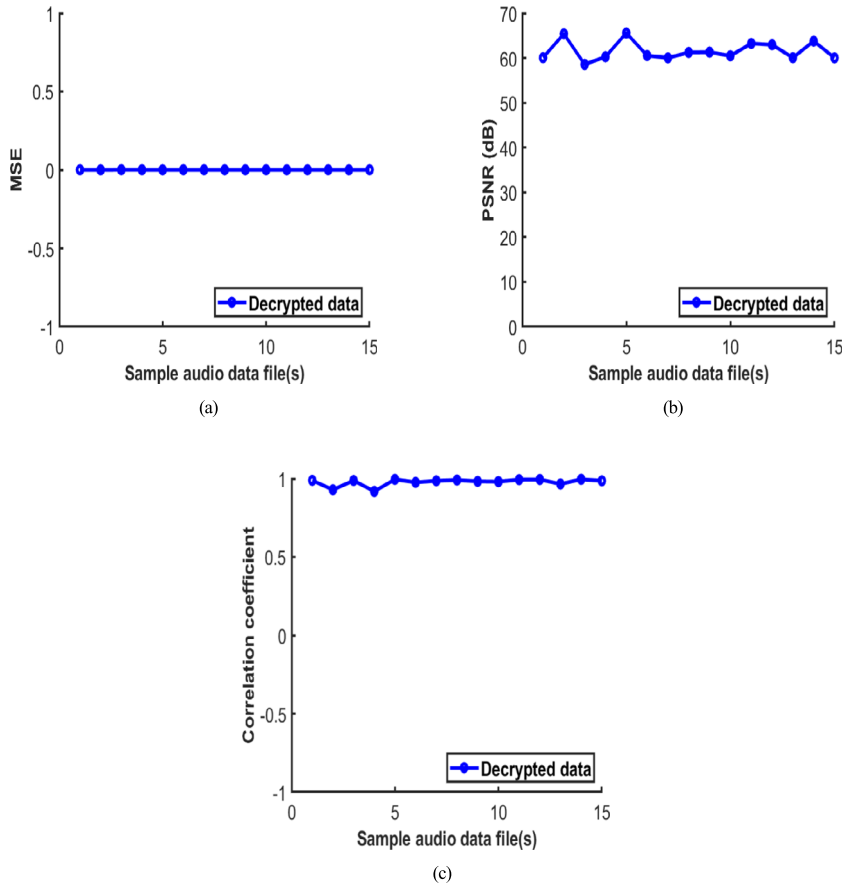


Fig. 5 **a** MSE of recovered audio. **b** PSNR of recovered audio. **c** Correlation coefficient of recovered audio

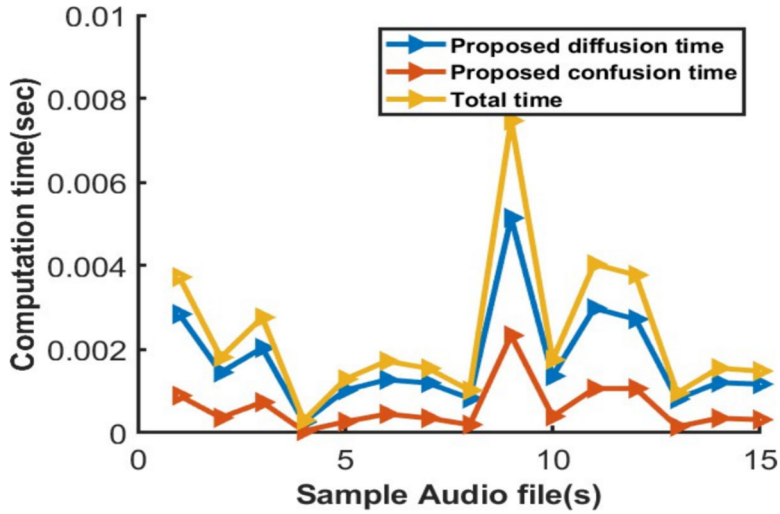


Fig. 6 Computational time indication for the individual steps in the proposed methodology

4 Comparison with existing works

4.1 Comparative analysis of the proposed encryption scheme

The correlation coefficient value, an indicator of randomness to adjacent amplitude values, is much closer to 0 in the case of the proposed work. Compared with the literature, the proposed encryption could achieve a correlation between adjacent values in the range of 10^{-5}. An entropy value closer to 14 indicates the complete random distribution of the amplitude values from − 1 to 1. The number of the sample changes rate and scrambling degree remained at the maximum values of 100 and 1, respectively, for any audio presented to the algorithm, as in Table 2. This demonstrates the encryption potential to transform confidential data into a statistically secure cryptogram entirely. The proposed technique outperforms the other existing methods considered for comparison. The audio is decrypted with a signal quality of 65.5804 dB PSNR and a correlation coefficient of 0.9952, as shown in Table 3.

4.2 Audio recovery performance analysis against existing works

Alongside augmented randomness, the technique is lossless, preserving the signal's quality. The proposed work could recover the audio with the very meagre MSE of 0.0005, closer to zero and a very good PSNR of 65.5804 dB, as in Table 3. The recovered signal seems almost the same as the transmitted signal, with the correlation coefficient closer to one. This infers that the proposed technique could recover the signal as well as the quality of signal masters in comparison with existing methods.

Table 2 Comparison of proposed encryption strength with existing works

Scheme	Correlation	Entropy	NSCR
Proposed method	− 0.000018	13.7553	100
Ref [14]	0.0013	NC	99.9994
Ref [21]	0.0010	NC	NC
Ref [28]	0.000013	NC	99.9999
Ref [37]	0.0001	4.3514	100
Ref [38]	-0.00054	NC	99.9978
Ref [33]	− 0.0635	7.8105	100
Ref [31]	0.00110	NC	NC
Ref [32]	0.0008	NC	99.71
Ref [34]	− 0.0005	7.9959	NC
Ref [46]	0.0223	NC	99.9997
Ref [45]	0.0002	NC	99.9932
Ref [18]	− 0.0094	NC	NC

NC not considered by the authors

Table 3 Audio recovery performance comparison with existing works

Scheme	MSE	PSNR	Correlation
Proposed method	0.0005	65.5804	0.9952
Ref [14]	0.04721	25.5492	NC
Ref [17]	0.212	52.67	NC
Ref [13]	NC	NC	0.98
Ref [25]	NC	49.877	0.9929
Ref [31]	NC	48.1931	NC
Ref [38]	-	26.0409	0.9950
Ref [12]	NC	60.08	NC

NC not considered by the authors

4.3 Comparative analysis of computational time

Provided that the proposed technique demonstrates the assured secrecy, efficiency, and reliability of the proposed system, the technique's viability on the current battery-supported wireless networks ought to be examined. The total computation time of the sender in the average falls at 5 ms, which is much less than the existing techniques as in Fig. 7. Thus, we could achieve computational time savings of 77.16% and 99.16% to [16] and [28]. The decryption is the exact reversal process that exhibits similar time complexity. The constant time consumption observed for the proposed technique assures resistance against time analysis attacks. This is common in the case of hardware-level attacks by adversaries.

4.4 Comparative key strength analysis

The proposed method has a wider keyspace of 2^{997} or 10^{300} than existing voice encryption schemes considered in Table 4. Though the key space is infinite in the range of (0,1), the key space is restricted to 10–15 for effective operation. Further, the encryption is

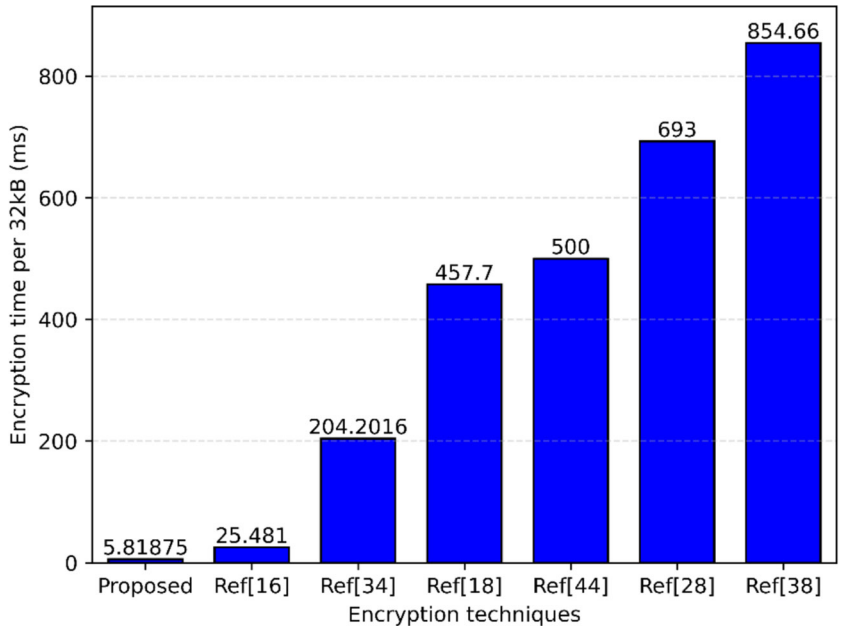


Fig. 7 Computational time comparison of the proposed methodology

Table 4 Key strength comparison with existing works

Scheme	Keyspace	Key sensitivity
Proposed Method	2^{997}	10^{-15}
Ref [19]	2^{605}	10^{-14}
Ref [29]	2^{128}	NC
Ref [25]	NC	10^{-4}
Ref [31]	2^{600}	NC
Ref [34]	2^{232}	NC
Ref [29]	NC	NC

NC not considered by the authors

responsive even to slight modifications in the range of 10^{-15}. The proposed adaptive key generation outputs a real number stream without iterative training. As long as the existing techniques predominantly generate the binary key stream, the FIS-based key generation could still pass the significant NIST tests. The proposed work thus proposes an intelligent AI-assisted cryptosystem. This encompasses user data-based FIS-based key generation and AIFES to defend against the robust security threats in wireless networks.

5 Conclusion

The multimedia data was vital for transmission over the present-day wireless networks. This comprises resource-constrained nodes exposed to an uncertain open world. The proposed work addresses these objectives by incorporating novel AI-dominated methodologies such as FIS-assisted key generation and lightweight AIFES-based fuzzy logic diffusion. The audio data of divergent sizes and distribution were considered. Following that, audio data is encrypted by fuzzy composition-based diffusion and chaotic confusion using the logistic map. Thus, the methodology could achieve the utmost security against various state-of-the-art attacks by employing rapid learning capability. This could be achieved through the constant noisy output with a spatial average correlation of − 0.00018 and entropy of 13.7553. The secure key exploiting attacks such as the Brute force attack shall be defended effectively. This could be achieved through identity preservation via an adaptive key generation scheme and eliminating public key exchange. Further, data sniffing, such as man-in-the-middle attacks, could be eradicated through AIFES through fuzzy diffusion. The adaptive feature selection mechanisms will be experimented with in future work to preprocess and choose more discriminative features for key generation depending on the data type.

Author contributions Venkatesan C and Lakshmi C designed the study, conducted the experiments, and analysed the data. Rebekka B. and Rengarajan Amirtharajan provided critical

feedback, helped write and edit the manuscript, and gave guidance throughout the project. All authors reviewed and approved the final manuscript.

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Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This article does not contain any studies with human participants performed by authors.

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